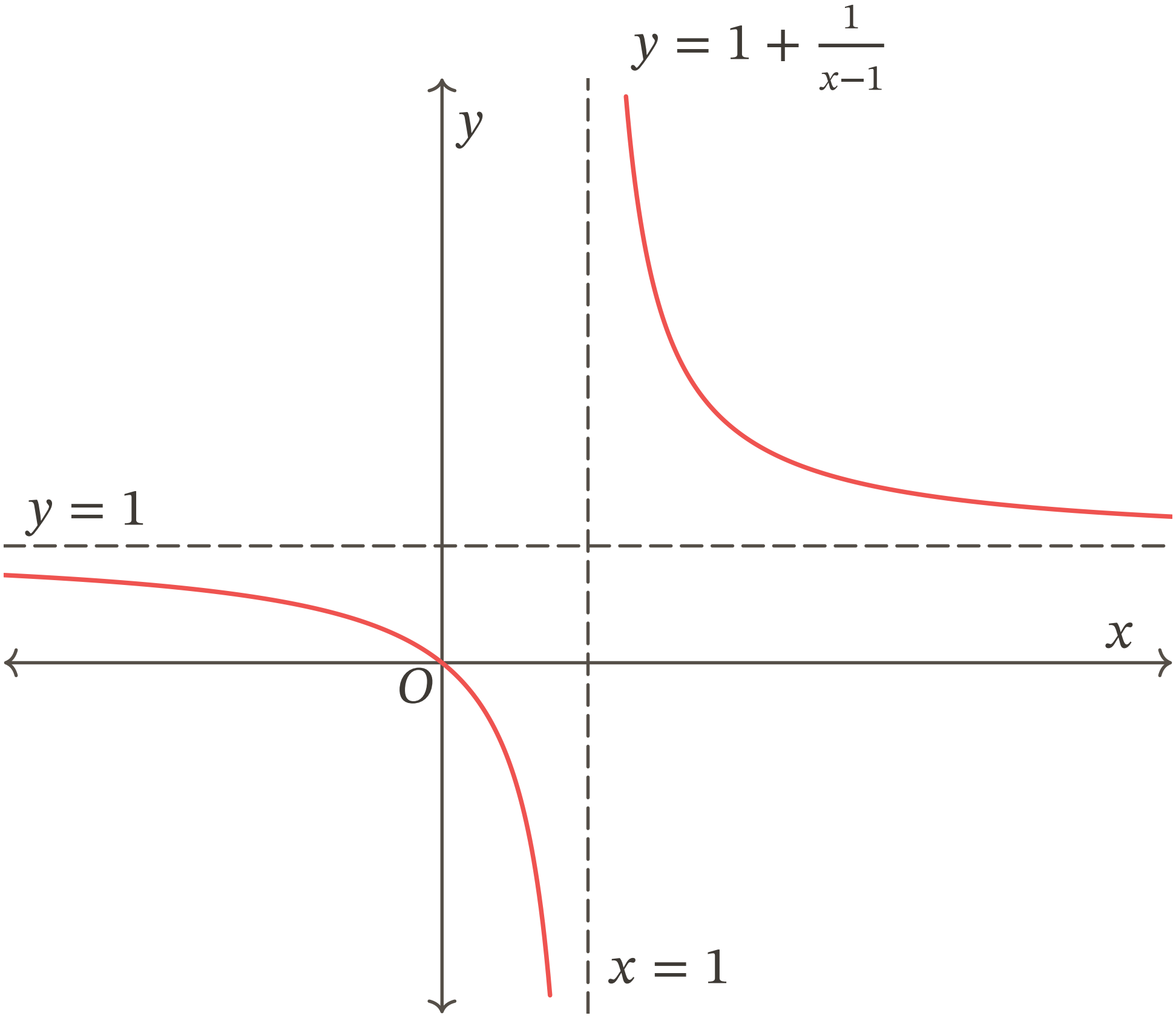
$y = 1 + \frac{1}{x-1}$
y
$y = 1$
x
O
$x = 1$